

Requirements – Movie Booking & Seat Booking System

1. Functional Requirements

Customer Details – Add and manage customer name, mobile number and other basic details.

Movie Details – Add and display movie name, language, duration and other movie information.

Theatre Details – Store and display theatre name, location and screen details.

Show Details – Manage show date, time and movie-theatre information.

Seat Availability – Display available and booked seats.

Seat Selection – Allow the customer to select available seats.

Seat Booking – Book the selected seats and store booking details.

Booking Confirmation – Generate and display confirmation after successful booking.

2. Non-Functional Requirements

Easy to use – Simple and user-friendly interface.

Performance – System should respond quickly.

Security – Customer and booking data should be protected.

Reliability – Booking information should be stored correctly.

Accuracy – Avoid duplicate seat bookings.

Maintainability – System should be easy to update and maintain.

3. Hardware Requirements

Computer/Laptop

Minimum 4 GB RAM

Minimum 20 GB free storage

Keyboard and Mouse

Internet connection (if required)

4. Software Requirements

Operating System: Windows/Linux

Frontend: HTML, CSS, JavaScript

Backend: PHP / Java / Python

Database: MySQL

IDE: VS Code / Eclipse / any suitable IDE

Web Browser: Chrome / Edge / Firefox

Algorithm – Movie Booking & Seat Booking System

Start

Enter Customer Details.

Select Movie.

Select Theatre.

Select Show Date and Time.

Display Available Seats.

Select the required Seat(s).

Check whether the selected seat is available.

If the seat is not available, select another seat.

If the seat is available, confirm the selected seat.

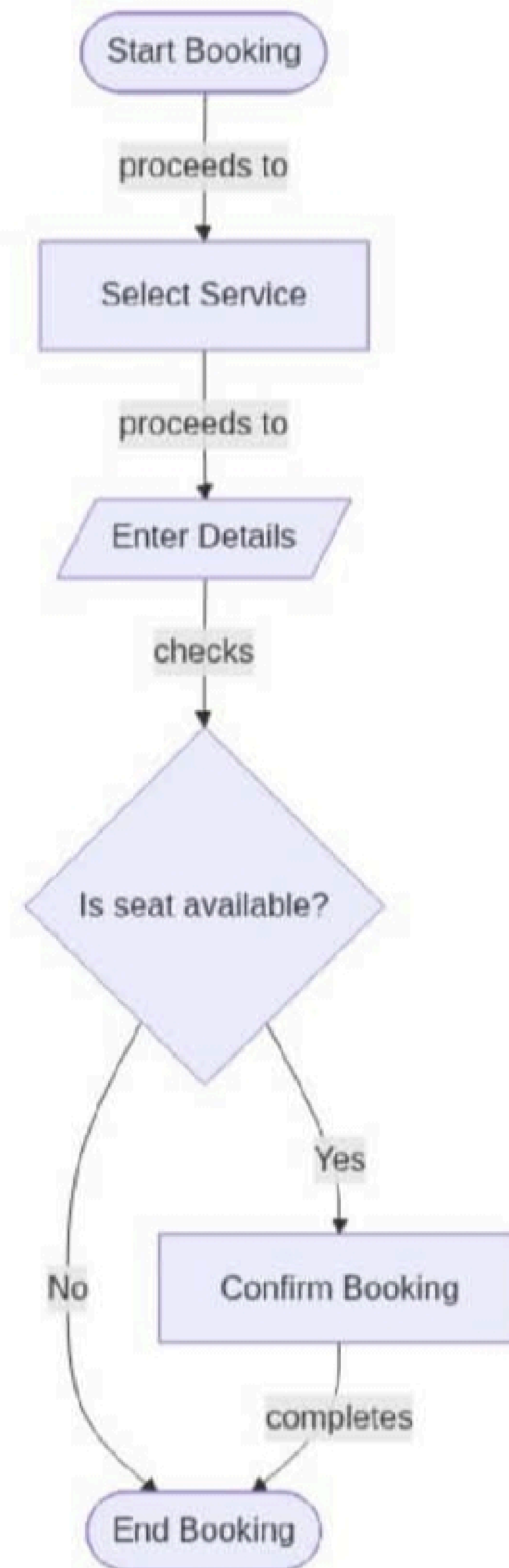
Calculate the Ticket Amount.

Confirm the Booking.

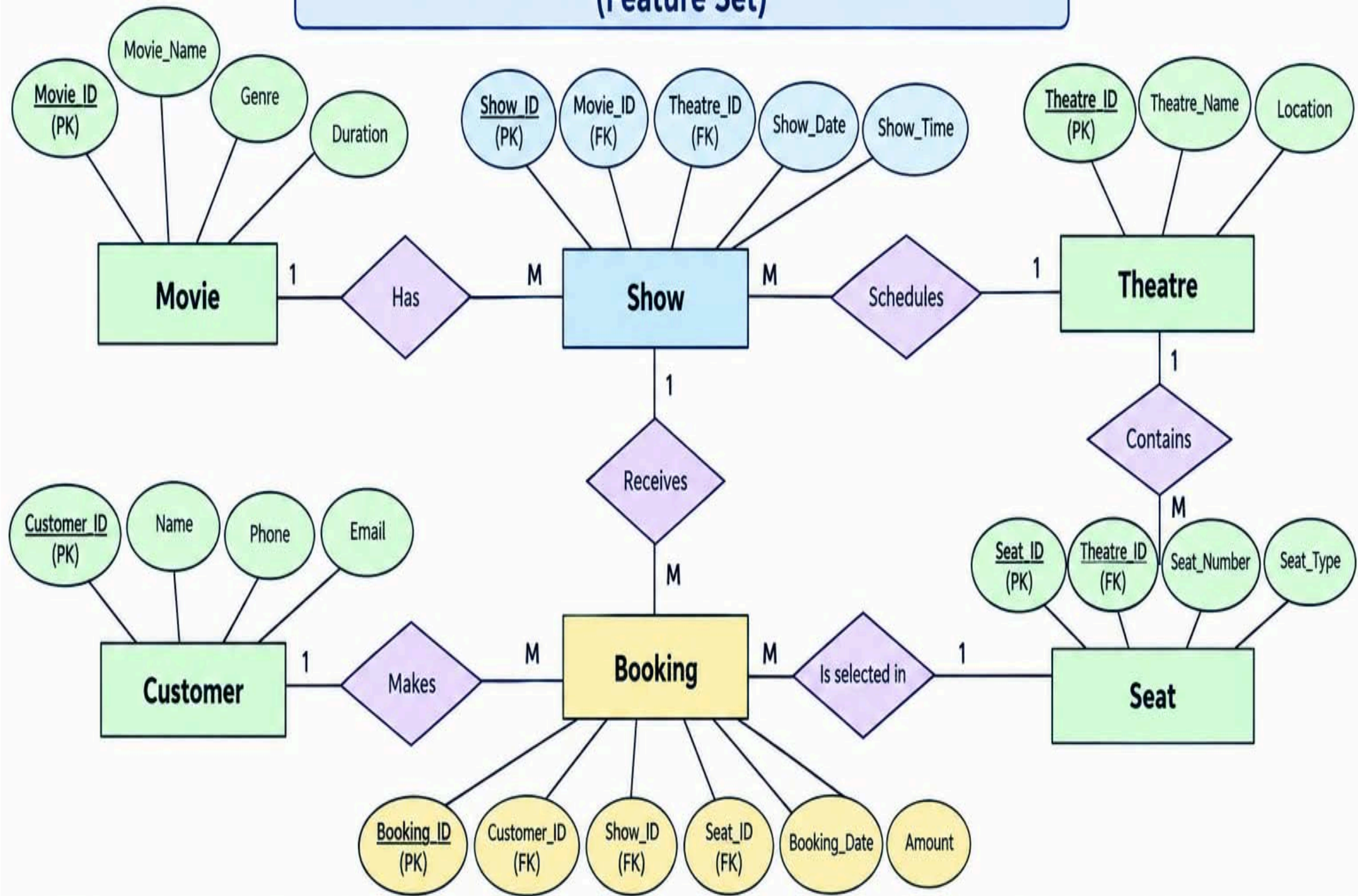
Display Booking Confirmation.

Stop.

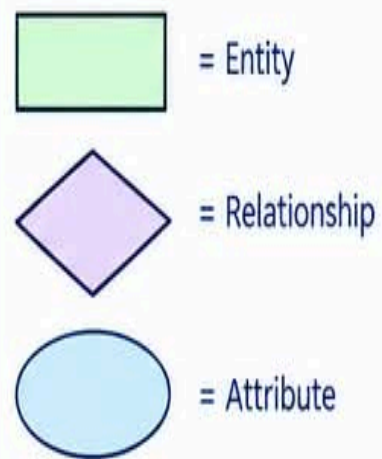
Complete Booking Process



ER Diagram - Movie Booking System (Feature Set)



Legend



PK = Primary Key
(underlined)
FK = Foreign Key
(marked as FK)
1:M = One to Many
1:1 = One to One

Relationships & Cardinality

- | | | | |
|-------------|-----------|---------|--|
| 1. Customer | → Booking | (1 : M) | (One customer can make many bookings, but each booking is made by one customer) |
| 2. Movie | → Show | (1 : M) | (One movie can have many shows, but each show is for one movie) |
| 3. Theatre | → Show | (1 : M) | (One theatre can schedule many shows, but each show is scheduled in one theatre) |
| 4. Theatre | → Seat | (1 : M) | (One theatre contains many seats, but each seat belongs to one theatre) |
| 5. Show | → Booking | (1 : M) | (One show can receive many bookings, but each booking is for one show) |
| 6. Seat | → Booking | (1 : M) | (One seat can be selected in many bookings, but each booking is for one seat) |

